

Minimum-Container Packing of a Non-Convex Polygon via Simulated Annealing and Bracketing–Bisection Search

Applied Mathematics / Numerical Methods — Graduate Report

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Abstract

We study the problem of packing N congruent non-convex polygons into a square of side length L and finding the minimum feasible L . Each polygon is a fixed 15-vertex shape, free to translate and rotate in the plane. We formulate packing as a continuous non-linear optimisation problem and solve it with a two-phase simulated annealing (SA) metaheuristic combined with an outer bracketing–bisection loop that certifiably narrows the interval containing the optimal L . Collision detection uses the Separating Axis Theorem (SAT) on a fan triangulation of the polygon, accelerated by a uniform spatial-hash grid and three-level axis-aligned bounding box (AABB) pruning. We report results for $N \in \{20, 50, 100, 200\}$, obtained on a SLURM cluster with 4–8 h wall-time jobs.

1 Problem Formulation

Let $\mathcal{P} \subset \mathbb{R}^2$ be a fixed non-convex polygon with vertices $\{v_k\}_{k=0}^{14}$ (15 vertices, unit-normalised coordinates). A placement of copy i is a rigid-body transformation $(c_i, \theta_i) \in \mathbb{R}^2 \times [0, 2\pi)$, giving world vertices $w_k^{(i)} = R(\theta_i)v_k + c_i$, where $R(\theta)$ is a 2×2 rotation matrix.

The *minimum-side packing* problem is:

$$L^* = \min_{L, \{(c_i, \theta_i)\}} L \quad \text{s.t.} \quad \begin{cases} \mathcal{P}_i \cap \mathcal{P}_j = \emptyset, & i \neq j, \\ \mathcal{P}_i \subset [-L/2, L/2]^2, & \forall i. \end{cases} \quad (1)$$

Area lower bound. Since $\sum_i \text{Area}(\mathcal{P}_i) \leq L^2$, an immediate necessary condition is

$$L \geq L_{\text{area}} := \sqrt{N \cdot A_p}, \quad (2)$$

where $A_p \approx 0.2456$ is the polygon area (shoelace formula). This bound is used throughout the search to prune provably infeasible candidates.

2 Penalty Reformulation

Problem (1) is handled via a penalty-based continuous relaxation. Define the *overlap depth* $d(T_a, T_b)$ between two triangles as the minimum SAT penetration depth (Section 3), and the *outside penalty* for polygon i as the squared AABB violation $\phi_i(L) = \sum_{\text{side}} (\text{violation})^2$. The energy is

$$E = \alpha_L \cdot L + \lambda \sum_{i < j} \sum_{a \in \mathcal{T}_i, b \in \mathcal{T}_j} d(T_a^{(i)}, T_b^{(j)})^p + \mu \sum_i \phi_i(L), \quad (3)$$

where $\mathcal{T}_i$ is the triangle fan of copy i , $p \in \{1, 2\}$, and (α_L, λ, μ) are weight schedules tuned per phase. Feasibility is declared when $\sum_{i < j} d_{ij} + \sum_i \phi_i < \epsilon_{\text{feas}} = 10^{-10}$.

3 Collision Detection

Polygon triangulation. Each 15-vertex polygon is decomposed off-line into $T = 13$ triangles by a fan from vertex 0, yielding three contiguous groups of (5, 5, 3) triangles. This enables three-level AABB culling: (1) polygon bounding circle, (2) per-group AABB, (3) per-triangle AABB, before invoking SAT.

SAT penetration depth. For two triangles A, B , we test all six edge-normal axes (three per triangle). On each axis $\hat{n}$, project both triangles and compute overlap $o = \min(\max A, \max B) - \max(\min A, \min B)$. If any $o \leq 0$, the triangles are separated. Otherwise $d(A, B) = \min_{\hat{n}} o > 0$ (penetration depth). The inner SAT loop operates directly on the world-vertex index arrays, avoiding per-iteration heap allocation.

Spatial hash grid. Polygon centres are stored in a uniform grid with cell size $h = 2r_{\text{br}}$ (twice the bounding radius). Neighbour search for copy k touches only cells within $R = \lceil 2r_{\text{br}}/h \rceil + 1$ of k 's cell, reducing average pair evaluations from $O(N^2)$ to $O(N)$. Incremental updates to the linked-list grid cost $O(1)$ per move.

4 Simulated Annealing

Move proposal. Each iteration picks a random copy k and applies one of: (i) local perturbation: $c_k += \delta_{xy} \sim U[-s_{xy}, s_{xy}]^2$, optional rotation $\theta_k += \delta_\theta \sim U[-s_\theta, s_\theta]$ (probability p_{rot}); or (ii) random reinsertion anywhere in $[-L/2, L/2]^2$ (probability p_{ri}). The energy change ΔE is computed *incrementally*: only the pairwise terms involving k are recomputed via the grid, reducing cost to $O(N_{\text{nbr}} T^2)$ per move.

Acceptance and adaptive step. Moves are accepted by the Metropolis criterion $\exp(-\Delta E/T)$. Step sizes (s_{xy}, s_θ) are adapted every $w = 1500$ iterations: shrunk by factor 0.88 if acceptance rate $< 25\%$, grown by 1.12 if $> 55\%$. A reheating mechanism restores $T \leftarrow \min(T \cdot r_h, T_{\text{max}})$ when acceptance drops below 2%.

Two-phase schedule. Each *trial* consists of two sequential phases:

	Phase A (explore)	Phase B (enforce)
Iterations	70 000	35 000
T_{start}	0.30	0.06
T_{end}	3×10^{-4}	10^{-6}
λ_0, μ_0	10^2	10^3 (ramped)
Overlap power p	1	2
p_{ri}	2%	0.5%

Phase A uses a linear overlap penalty ($p = 1$) with large steps to explore configuration space. Phase B switches to quadratic overlap ($p = 2$) with multiplicative penalty ramping ($\times 1.8$ every 1200 iterations) to enforce strict feasibility. The temperature schedule is geometric: $T_t = T_0 \beta^t$, $\beta = (T_{\text{end}}/T_{\text{start}})^{1/N_{\text{iter}}}$.

Deterministic HPC seeding. Each trial seed is derived as $s_{\text{trial}} = f(\text{base_seed}, \text{run_id}, \text{trial_id})$ using Split-Mix64 mixing, ensuring full reproducibility across SLURM array jobs without time-based reseeding. The PRNG is xorshift64*.

5 Outer Minimization: Bracketing, Bisection, and Polish

The outer loop maintains an interval $[L_{\text{low}}, L_{\text{high}}]$ with L_{low} infeasible and L_{high} feasible, then narrows it by bisection, and finally conducts an adaptive long-run polish. Algorithm 1 summarises the procedure.

Algorithm 1 Outer minimisation

- 1: Initialise L from grid layout; $L_{\text{low}} \leftarrow L_{\text{area}}$
 - 2: **Bracket**: shrink/grow L by 3%/trial until both feasible and infeasible endpoints are found
 - 3: **Bisect**: 26 steps of $L_{\text{mid}} \leftarrow (L_{\text{low}} + L_{\text{high}})/2$; run K_b SA trials; update bracket
 - 4: **Polish**: adaptive shrink $L \leftarrow L^*(1 - \varepsilon)$; accept if feasible; adapt $\varepsilon \in [10^{-5}, 2 \times 10^{-3}]$ to target 35% success rate over windows of 20 attempts
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The warm-start strategy scales existing positions by $\gamma = (L_{\text{new}}/L_{\text{old}}) \cdot 0.98$ before each SA run, preserving structural information across successive values of L . A global best snapshot is flushed to disk on SIGTERM for safe SLURM preemption.

6 Experimental Results

Setup. Experiments ran on the University of Arizona HPC cluster (SLURM). $N \in \{20, 50, 100\}$ used 4 h wall-time; $N = 200$ used 8 h. The bracketing phase used $K_{\text{br}} = 10$ trials per candidate L ; bisection used $K_{\text{bs}} = 7$; polish used $K_p = 5$. Results are reported as the best L achieving full feasibility ($\text{feas} < 10^{-10}$).

Table 1: Best packing side length L^* and area efficiency $\eta = NA_p/(L^*)^2$.

N	L_{area}	L^*	η	Wall time
20	2.216	(see fig.)	—	4 h
50	3.504	(see fig.)	—	4 h
100	4.956	(see fig.)	—	4 h
200	7.009	(see fig.)	—	8 h

Fill in L^* and $\eta = NA_p/(L^*)^2$ from your `csv/*_best_polys_N***.csv` header after your jobs complete. $\eta = 1$ would match the area lower bound; typical packing densities for non-convex shapes are 0.75–0.95.

7 Conclusion and Future Work

We presented a complete solver for minimum-container polygon packing, combining two-phase SA with bracketing–bisection search and an adaptive polish loop. The SAT-based collision engine with grid acceleration and multi-level AABB culling scales the inner loop to practical N , and deterministic SLURM-safe seeding ensures reproducibility.

Future directions. Three natural extensions suggest themselves. First, *GPU acceleration*: the outer $O(N^2)$ pair-penalty computation and the embarrassingly parallel



Figure 1: Best feasible packings for $N = 20, 50, 100, 200$ (left-to-right, top-to-bottom). Each polygon is a 15-vertex non-convex shape rendered in the square $[-L^*/2, L^*/2]^2$.

multi-trial structure map naturally to CUDA, with per-block triangle pairs and shared-memory polygon tiles; preliminary roadmaps target a Tesla P100. Second, *SA variant comparison*: parallel tempering (PT) and event-driven multicanonical sampling (ER-MS) offer theoretical mixing-time advantages over standard SA and merit a controlled benchmark. Third, *generalised polygon families*: the triangulation and SAT kernel are geometry-agnostic and can accommodate arbitrary convex or star-shaped polygons with minor preprocessing changes.

References

- [1] S. Kirkpatrick, C. D. Gelatt, and M. P. Vecchi, “Optimization by Simulated Annealing,” *Science*, 220(4598):671–680, 1983.
- [2] A. M. Sutton, A. E. Howe, and L. D. Whitley, “A Theoretical Analysis of the $(1 + 1)$ EA on ± 1 -Linear Functions,” *GECCO*, 2012.
- [3] J. Egeblad, B. K. Nielsen, and A. Odgaard, “Fast neighbourhood search for two- and three-dimensional nesting problems,” *European Journal of Operational Research*, 197(2):543–559, 2009.
- [4] C. M. Gomes and R. Shmoys, “Completing quasi-groups or Latin squares: A structured graph coloring problem,” *Annals of OR*, 2006.